Optimal histogram binning for EMCCD photon-counting flux retrieval

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How many bins, and exactly where to cut
Generated by `bin_optimizer.py` and `demo_optimal_bins.py`
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Answer in one box

For an EMCCD read out with bias b, mean EM gain G, read noise σ and saturation at x_{sat}, working over $\mu \leq 8$ e$^-$ per read:

Eight bins are enough. They retain $\geq 98.3\,\%$ of the accuracy of the full, undegraded 1 ADU histogram at every flux from 10^{-3} to 8 e$^-$/read. Four bins do not (89.5 %); eight is the smallest power of two that clears the 95 % target, with margin.

Where the cuts go. Write the pixel value in gain units, $u = (x - b)/G$, and let $S = (x_{\mathrm{sat}} - b)/G$.

1. One cut just above the read-noise peak, at $x = b + c_1\sigma$ with $c_1 \simeq 4$ to 4.5. This one number does almost all the work at low flux.
2. Six rungs spaced *uniformly in* $\sqrt{u}$, from $u_1 \simeq 1$ (one electron) up to $u_{\mathrm{top}} = \min(S,\ \mu_{\max} + 3\sqrt{\mu_{\max}})$.
3. One open bin above u_{top}, absorbing saturation and overflow.

The wide bin between cut 1 and u_1 is the single-electron shoulder and is deliberately left unsplit.

1 The question, and what "95 % of the accuracy" means

An EMCCD flux estimator that never stores the frames keeps, for each pixel, a histogram of that pixel's raw ADU values across all reads, and fits the mean flux μ from that histogram. The histogram is the entire memory footprint of the method, so its number of bins K is the parameter that matters. The question is how small K can be before the flux estimate measurably degrades, and where the K bin edges belong.

The detector digitises to whole ADU, so there is a natural, finite reference: the *full histogram*, one bin per ADU,

$$\text{cell } i = [i,\ i+1) \ \text{ for } i < x_{\mathrm{sat}}, \qquad \text{cell } x_{\mathrm{sat}} = [x_{\mathrm{sat}},\ \infty), \tag{1}$$

the last cell holding everything that saturates, since saturated pixels are mutually indistinguishable. Any K-bin histogram is a merge of contiguous runs of these cells. By the data-processing inequality it can therefore never carry *more* information than the full histogram, which makes the comparison below a clean, bounded one.

The relevant measure is the Fisher information about μ carried by one read of one pixel,

$$I(\mu) \;=\; \sum_{\text{bins } i} \frac{1}{p_i(\mu)} \left(\frac{\partial p_i}{\partial \mu}\right)^2, \tag{2}$$

with $p_i(\mu)$ the probability that a read lands in bin i. Over N reads the Cramér-Rao bound gives $\sigma_{\hat{\mu}} = 1/\sqrt{NI(\mu)}$, so the comparison between a K-bin design and the full histogram is

$$\boxed{\eta(\mu) \;=\; \frac{I_K(\mu)}{I_{\text{full}}(\mu)}, \qquad \frac{\sigma_{\text{full}}}{\sigma_K} \;=\; \sqrt{\eta(\mu)}} \tag{3}$$

We call η the *information efficiency* and $\sqrt{\eta}$ the *accuracy retained.* "Recovering more than 95 % of the accuracy" is read literally as $\sqrt{\eta} \geq 0.95$ at every flux in the range of interest, i.e. $\eta \geq 0.9025$. Both numbers are quoted throughout, because the stricter reading $\eta \geq 0.95$ is also of interest and the eight-bin design happens to satisfy it too.

Two properties of (2) drive everything that follows. It is a *sum over bins*, which makes the optimisation exactly solvable (Section 4); and it is bounded above by the full-histogram value, which makes $\eta \leq 1$ by construction.

2 The signal chain, and its two dimensionless numbers

One read of one pixel is

$$n \sim \text{Poisson}(\mu + \mu_{\text{CIC}}), \qquad g \sim \text{Gamma}(n, G) \;\; (g = 0 \text{ if } n = 0), \qquad x = b + g + \mathcal{N}(0, \sigma^2), \tag{4}$$

truncated at x_{sat}. The zero-electron case is a pure Gaussian at the bias; each additional electron adds one more Gamma stage of mean G. This is the standard EMCCD model of Basden et al. (2003), as used by Harpsøe et al. (2012) and Daigle et al. (2009).

Rescaling by the gain removes all but two free numbers:

$$u = \frac{x - b}{G}, \qquad r = \frac{\sigma}{G} \text{ (read noise in gain units)}, \qquad S = \frac{x_{\text{sat}} - b}{G} \text{ (saturation head-room).} \tag{5}$$

The optimal design depends on the detector only through r and S, and on the science only through the flux range $[\mu_{\min}, \mu_{\max}]$. The worked example throughout is $b = 1000$, $G = 5000$ ADU/e$^-$, $\sigma = 30$ ADU, $x_{\text{sat}} = 60000$ ADU, $\mu_{\text{CIC}} = 10^{-3}$, giving $r = 0.0060$ and $S = 11.8$ e$^-$. A second detector with a much larger r is treated in Section 8.

3 Where the flux information actually sits

Before optimising anything, it pays to see what the answer must look like. Figure 1 shows the pixel-value distribution and the Fisher information density, split into the read-noise peak (left, in units of σ) and the EM tail (right, in units of G). These are two utterly different scales: the peak is $\sim 10^{-2}$ wide in u, the tail is ~ 10 wide.

3.1 Low flux: the histogram collapses to a single threshold

Consider $\mu \to 0$, so that reads are almost all $n = 0$ with an occasional $n = 1$. Put the first cut at u_1, and suppose the read noise leaks negligibly above it. Then every bin j lying above the cut contains only single-electron events, so $p_j = \lambda g_j$ and $\partial p_j / \partial \mu = g_j$, where g_j is the $n = 1$ mass in that bin and $\lambda = \mu + \mu_{\text{CIC}}$. Its contribution to (2) is

$$\frac{(\partial p_j / \partial \mu)^2}{p_j} \;=\; \frac{g_j}{\lambda}, \qquad \text{so} \qquad \sum_{j \text{ above the cut}} \frac{g_j}{\lambda} \;=\; \frac{e^{-u_1}}{\lambda}. \tag{6}$$

The sum *telescopes.* It depends on where the first cut is and on nothing else: how the region above the cut is subdivided is irrelevant. At low flux the histogram is worth exactly as much as a single threshold.

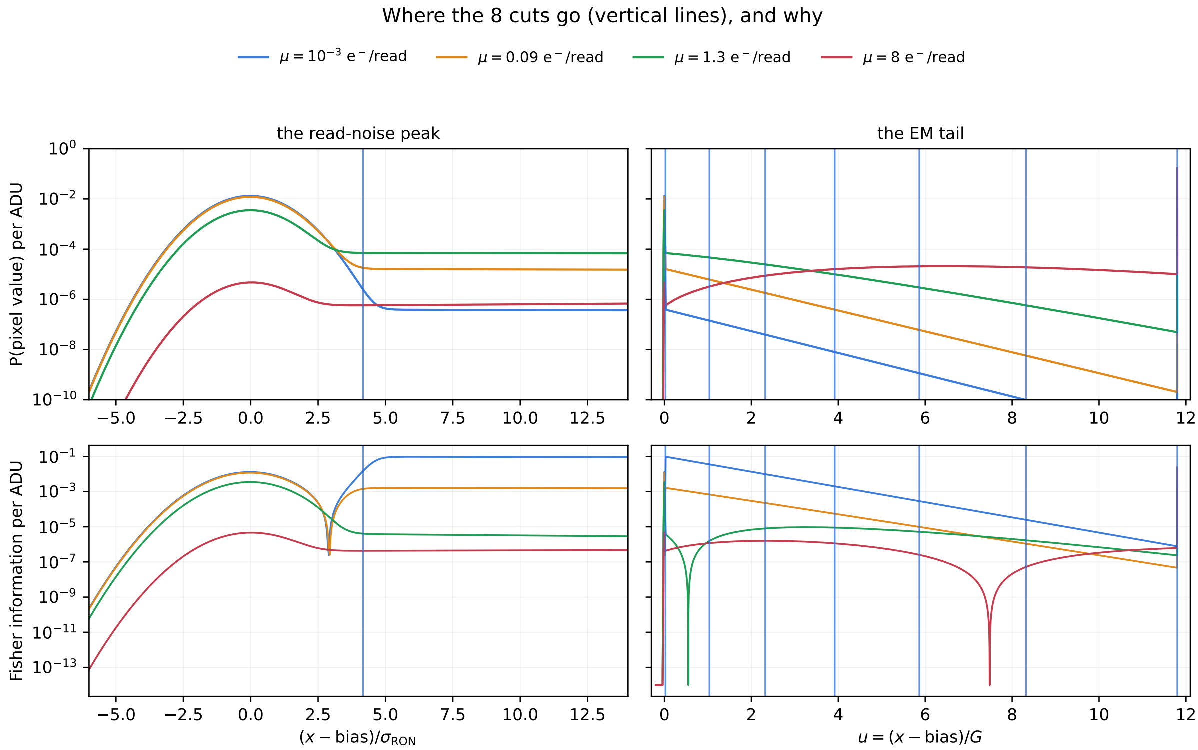

Figure 1: The eight cuts (vertical lines) against the distribution they bin. Left: the read-noise peak, in units of the read noise. Right: the EM tail, in units of the gain. Top: the pixel-value probability per ADU. Bottom: the Fisher information about μ per ADU. A single cut at $+4.2\sigma$ serves the whole peak; the remaining cuts climb the tail. The dips in the bottom panels are where $\partial p/\partial\mu$ changes sign, not zeros of the density.

That is not a heuristic, and Table 1 confirms it numerically. Two bins, meaning one threshold, already match the full eight-bin design below $\mu \sim 10^{-2}$ e$^-$/read. Bins only begin to earn their keep as μ approaches one electron per read, when double and triple electrons populate the tail and its *shape*, not just its total, starts to carry information.

Table 1: Information efficiency η of designs sharing the same first cut (1125 ADU) but differing in how many bins sit above it. At low flux the extra bins buy nothing, exactly as (6) predicts.

	μ [e$^-$/read]				
design	10^{-3}	10^{-2}	10^{-1}	1	8
2 bins (one threshold)	0.9944	0.9985	0.9940	0.8199	0.0062
3 bins	0.9944	0.9985	0.9948	0.8983	0.4411
8 bins, optimal	0.9945	0.9985	0.9963	0.9827	0.9678

3.2 The first cut: a tension between two leaks

Equation (6) assumed the read noise does not leak above the cut. That assumption is the single most important design constraint, and getting it wrong is catastrophic rather than merely suboptimal. If a fraction $Q(c_1) = \frac{1}{2}\,\text{erfc}(c_1/\sqrt{2})$ of the zero-electron peak sits above a cut at $b + c_1\sigma$, then the bin just above the cut contains a flux-independent population Q alongside the flux-carrying population $\sim \lambda$. Its contribution stops scaling as $1/\lambda$ and saturates at $O(1)$: the information is gone.

So c_1 is squeezed from both sides:

$$\text{from below:} \quad Q(c_1) \ll \mu_{\min} \qquad \text{or the peak's own tail swamps the electrons;} \tag{7}$$

$$\text{from above:} \quad 1 - e^{-c_1 r} \ll 1 \qquad \text{since that fraction of single electrons falls into bin 0.} \tag{8}$$

Table 2 evaluates both. For the worked detector, $c_1 = 4$ to 4.5 is comfortable: the leak is 10^{-5} or below, well under $\mu_{\min} = 10^{-3}$, while only 2.4 to 2.7 % of single electrons are sacrificed. The numerical optimum lands at $c_1 = 4.17$, which is simply where these two curves cross.

This is also where the user's intuition about "a few sigma" is made precise: the right number of sigmas is $c_1 \simeq Q^{-1}(\mu_{\min})$, so a lower target flux pushes the cut outward, logarithmically slowly.

Table 2: The two competing leaks as a function of the first cut position c_1, for the worked detector ($r = 0.0060$) and for a detector with much larger read noise relative to gain ($r = 0.0959$, Section 8). The optimum sits where the read-noise leak has become negligible but the electron loss has not yet become expensive.

c_1 [σ]	2.0	3.0	3.5	4.0	4.5	5.0
read-noise leak $Q(c_1)$	$2.3{\cdot}10^{-2}$	$1.4{\cdot}10^{-3}$	$2.3{\cdot}10^{-4}$	$3.2{\cdot}10^{-5}$	$3.4{\cdot}10^{-6}$	$2.9{\cdot}10^{-7}$
electrons lost, $r = 0.0060$	1.19 %	1.78 %	2.08 %	2.37 %	2.66 %	2.96 %
electrons lost, $r = 0.0959$	17.5 %	25.0 %	28.5 %	31.9 %	35.0 %	38.1 %

3.3 The tail: uniform in $\sqrt{u}$

Above the peak the distribution is a Poisson mixture of Gamma laws. The n-electron component has mean nG and standard deviation $\sqrt{n}\,G$, so at a pixel value u the local width of the structure that must be resolved grows like $\sqrt{u}$. Bins whose widths follow that growth, i.e. rungs placed uniformly in $\sqrt{u}$, keep a constant number of bins per local width everywhere along the tail. Formally, $\sqrt{u}$ is the variance-stabilising transform of the Gamma family, the same argument that puts $\sqrt{\cdot}$ at the heart of Anscombe-type transforms.

The numerical optimum reproduces this without being told. For the eight-bin design the successive rungs sit at

$$\sqrt{u} \;=\; 1.02, \quad 1.52, \quad 1.98, \quad 2.42, \quad 2.88, \quad 3.44,$$

with increments 0.51, 0.46, 0.44, 0.46, 0.55: uniform to within a few percent, over a range in u itself spanning more than a factor of ten. Restricting the closed-form family of Section 4 to one ladder shape at a time and re-optimising every other parameter gives worst-case $\eta = 0.9643$ for $\sqrt{u}$, 0.9594 for $\log u$ and 0.9480 for u itself. The log-uniform ladder, the natural first guess, is the runner-up rather than a disaster; it over-resolves the region just above the peak, where (6) has already shown there is nothing to resolve, and pays for it at the top of the range.

3.4 The top cut: stop where the counts stop

The last finite edge should not be placed at saturation out of habit. Bins placed where no read ever lands are wasted. Since the pixel value of an n-electron event is $\sim n \pm \sqrt{n}$ in gain units, and n itself rarely exceeds $\mu + 3\sqrt{\mu}$,

$$u_{\text{top}} \;=\; \min\Big(S, \quad \mu_{\max} + 3\sqrt{\mu_{\max}}\Big). \tag{9}$$

For the worked detector $S = 11.8$ is smaller than $8 + 3\sqrt{8} = 16.5$, so real saturation binds and the top cut sits at x_{sat}. For a detector with generous head-room the second term binds instead, and the optimiser duly places the top cut well below saturation (Section 8).

Figure 2 closes the argument: it shows the cumulative fraction of the flux information as a function of pixel value. The cuts are placed so that each bin carries a comparable share of the information, which is what maximising (2) under a bin budget amounts to.

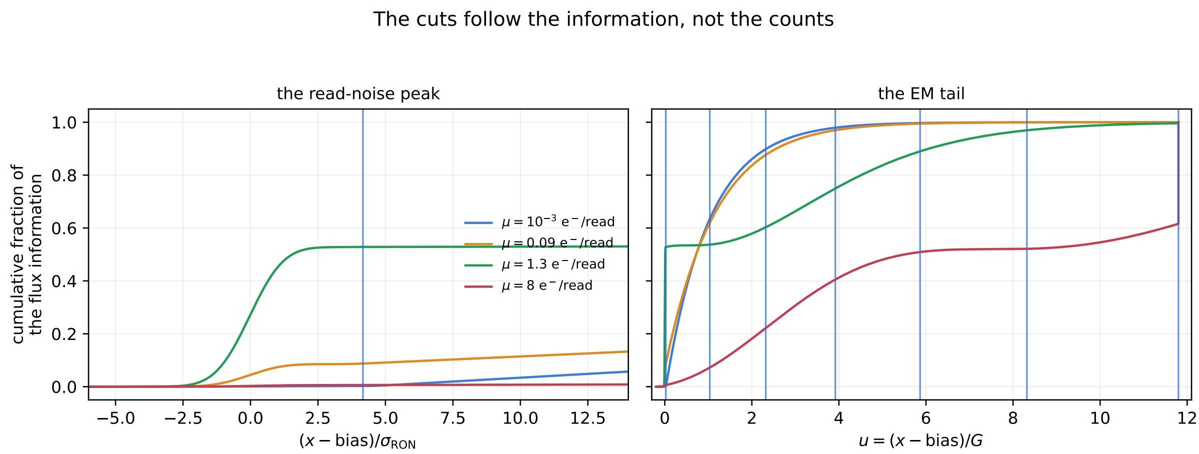

Figure 2: Cumulative fraction of the flux information versus pixel value, at four fluxes, with the eight cuts overlaid. At $\mu = 10^{-3}$ e$^-$/read essentially all of the information is collected by the first cut. By $\mu = 8$ e$^-$/read it is spread across the whole tail, which is why the tail rungs exist.

4 Finding the optimum, not just a good guess

Because (2) is additive over bins, a weighted objective

$$V = \sum_{\mu} w_{\mu} \frac{I_K(\mu)}{I_{\text{full}}(\mu)} = \sum_{\text{bins } i} \underbrace{\left[\sum_{\mu} \frac{w_{\mu}}{I_{\text{full}}(\mu)} \frac{(\Delta_i \partial_{\mu} P)^2}{\Delta_i P} \right]}_{\text{depends only on bin } i} \tag{10}$$

is also additive over bins, where Δ_i denotes the difference of the cumulative distribution across bin i. A contiguous partition with an additive value admits an exact dynamic program: the best K-bin partition ending at a given edge is built from the best $(K-1)$-bin partition ending at each earlier edge. Restricted to a dense candidate set of edges (about 560 positions, spaced 0.1σ through the peak and logarithmically above it), the optimum is therefore *global*, not the result of a local search.

The design must be good over a *range* of fluxes, not at one flux, so what is actually wanted is the max-min solution $\max_{\text{edges}} \min_{\mu} \eta(\mu)$. That is not additive, but it is the dual of a weighted problem: a multiplicative-weights loop repeatedly solves the additive problem exactly and then shifts w_{μ} towards whichever fluxes are currently worst served. Forty rounds converge to within 10^{-3} in η and take under two seconds.

The result is a hard yardstick. A closed-form family that matches it is not merely plausible, it is provably close to the best a K-bin histogram can do. The three-zone family of the summary box recovers 99.9 % of the dynamic programme's worst-case η at $K = 8$, and matches it at $K = 16$.

5 Results

5.1 How many bins

Table 3 and Figure 3 give the headline result. Eight bins is the smallest power of two that meets the target, and it does so with room to spare: 98.3 % of the full-histogram accuracy in the worst case, against a 95 % requirement. Six bins would also pass, but eight is both a power of two and noticeably safer. Four bins fall short at 89.5 %.

The two-bin entry is missing from Table 3 because it deserves a section of its own: two bins is one threshold, which is classical photon counting. Section 5.4 compares against it properly, giving thresholding the best cut it could possibly have.

Table 3: Worst-case performance over $\mu \in [10^{-3}, 8]$ e^-/read, for the worked detector. "Accuracy" is $\sigma_{\rm full}/\sigma_K = \sqrt{\eta}$, the quantity the 95 % target applies to. The last column is what K equal-width bins from bias to saturation would give.

K	η (worst)	accuracy	meets 95 %?	equal-width accuracy
4	0.8015	89.5 %	no	14.3 %
5	0.8859	94.1 %	no	23.4 %
6	0.9284	96.4 %	yes	31.5 %
7	0.9521	97.6 %	yes	38.3 %
8	**0.9657**	**98.3 %**	**yes**	44.1 %
12	0.9872	99.4 %	yes	59.8 %
16	0.9928	99.6 %	yes	69.0 %
32	0.9982	99.9 %	yes	78.4 %

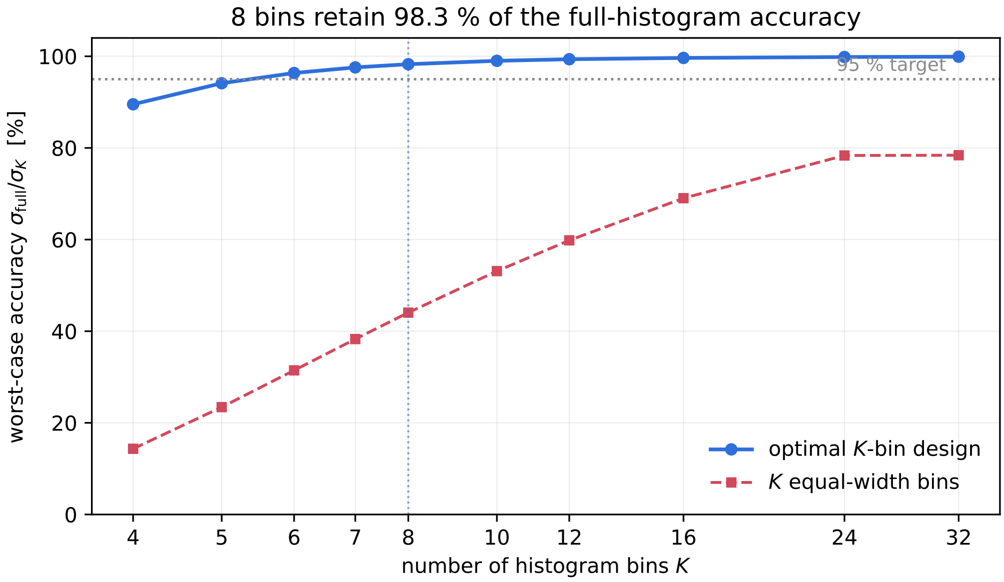

Figure 3: Worst-case accuracy retained versus the number of bins, over $\mu \in [10^{-3}, 8]$ e^-/read. The optimised design saturates quickly; equal-width bins never recover.

The equal-width column is worth dwelling on. Spreading K bins uniformly from bias to saturation, the obvious thing to do without thinking about it, gives 44 % at $K = 8$ and still only 78 % at $K = 32$: it never catches up, because it spends essentially all of its bins on the empty region above the signal and none on the read-noise peak, where at low flux all of the information is. The gain from thinking about bin placement is far larger than the gain from quadrupling the number of bins.

5.2 The eight-bin design, cut by cut

Table 4 lists the design. The first cut sits at 1125 ADU $= b + 4.17\sigma$; every later cut is at hundreds of σ above the bias and is set by the gain, not by the read noise. As an edge list in the convention used by `embin.py` and `emccd_histo.get_bins`, where the final entry is a formal sentinel for the open top bin:

```
edges: [0, 1125, 6176, 12596, 20591, 30324, 42595, 60000, 60001]
```

Table 4: The eight-bin design for $b = 1000$, $G = 5000$ ADU/e$^-$, $\sigma = 30$ ADU, $x_{sat} = 60000$ ADU, optimised over $\mu \in [10^{-3}, 8]$ e$^-$/read. Bin 0 absorbs everything below the first cut, bin 7 everything above the last. The rightmost column shows the cut positions in $\sqrt{u}$, whose near-constant spacing above u_1 is the signature discussed in Section 3.

bin	lower [ADU]	upper [ADU]	u_{lo}	u_{hi}	$\sqrt{u_{hi}}$
0	0	1 125	−0.200	0.025	0.16
1	1 125	6 176	0.025	1.035	1.02
2	6 176	12 596	1.035	2.319	1.52
3	12 596	20 591	2.319	3.918	1.98
4	20 591	30 324	3.918	5.865	2.42
5	30 324	42 595	5.865	8.319	2.88
6	42 595	60 000	8.319	11.800	3.44
7	60 000	∞	11.800	∞	

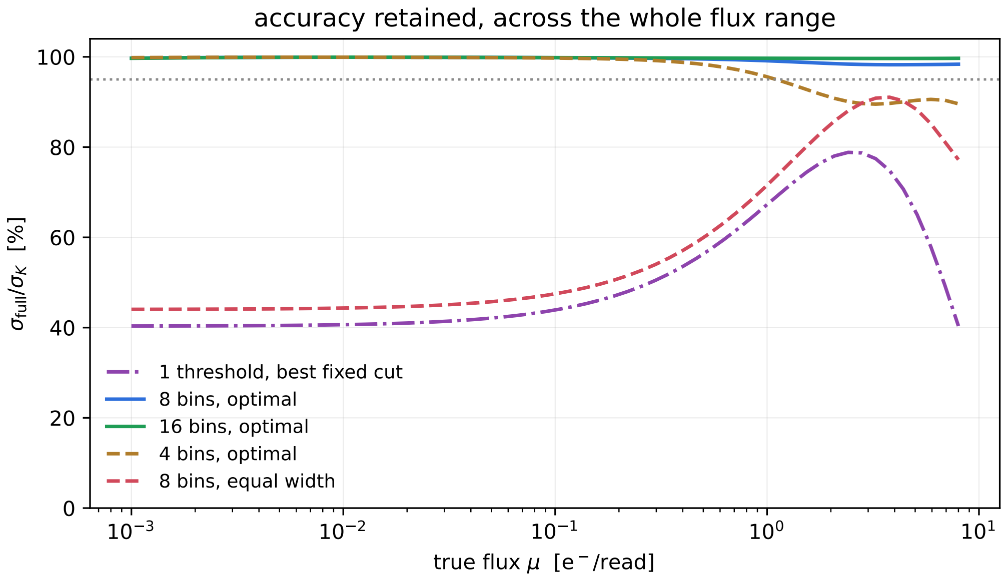

Figure 4: Accuracy retained versus true flux, for several designs. Four optimised bins fail only above $\mu \sim 1$ e$^-$/read; eight are flat at 98 to 100 % throughout. Eight equal-width bins lose more than half the accuracy at low flux. The single best-fixed-cut threshold is shown for scale only; Section 5.4 treats thresholding on fairer terms.

5.3 Efficiency across the flux range

Figure 4 shows $\sqrt{\eta}$ as a function of the true flux. The eight-bin design is essentially perfect below $\mu \sim 0.3$ e$^-$/read, where Section 3 showed a single threshold suffices, and dips to its worst value in the few-electron regime:

μ [e$^-$/read]	0.5	1	2	3	4	6	8
η	0.9906	0.9827	0.9708	0.9661	0.9653	0.9663	0.9678
accuracy	99.5 %	99.1 %	98.5 %	98.3 %	98.3 %	98.3 %	98.4 %

The binding constraint is at a few electrons per read, precisely the regime the design is meant to protect, and the curve is flat at 98.3 % from 3 e$^-$/read onwards. This is not an accident: the max-min optimisation places the worst case there deliberately, because that is where a fixed bin budget is hardest to spend well. Below one electron per read the problem is easy; above a few electrons per read the Gamma tail becomes broad and smooth, and coarse bins stop hurting.

5.4 Against optimal thresholding

Classical EMCCD photon counting keeps one bit per read: did this pixel exceed a threshold or not? In the language of this note that is the two-bin case, and Equation (6) already explained why it is nearly optimal when $\mu \ll 1$. What it does not say is where thresholding stops being nearly optimal, or how far it can be pushed by choosing the cut well rather than merely safely. Three thresholds are worth separating:

- **Photon counting.** One fixed cut at $b + 5\sigma$, placed to keep the read-noise peak out of the counted population and for no other reason. For the worked detector that is 1150 ADU.
- **Best fixed cut.** The single cut that maximises the *worst* efficiency over the whole flux range, i.e. thresholding tuned by exactly the max-min criterion the bin design is tuned by. It lands at 10 231 ADU $= b + 1.85\,G$, high in the EM tail.
- **Best cut at each flux.** The cut re-chosen for every μ. This needs the answer in advance, so it is not a method; it is a *ceiling* on the method. No thresholding scheme, however clever, sits above this curve.

Table 5: Accuracy $\sqrt{\eta}$ retained by one threshold, against the eight-bin design, for the worked detector. The last column is the ceiling: the best any single threshold could do at that flux, knowing the flux.

μ [e$^-$/read]	photon counting (5σ, 1150 ADU)	best fixed cut (10 231 ADU)	8 bins optimal	ceiling (best cut at this μ)
0.5	96.9 %	56.3 %	99.5 %	97.2 %
1	90.5 %	67.3 %	99.1 %	90.6 %
2	72.4 %	77.7 %	98.5 %	79.0 %
3	53.8 %	78.3 %	98.3 %	79.1 %
4	38.3 %	73.5 %	98.3 %	79.4 %
6	17.9 %	57.3 %	98.3 %	80.0 %
8	8.0 %	40.4 %	98.4 %	81.1 %
worst over $[10^{-3}, 8]$	8.0 %	40.4 %	**98.3 %**	78.9 %

Table 5 and Figure 5 say three separate things, and it is worth not blurring them together.

Below one electron per read, thresholding is not merely adequate, it is essentially optimal. The ceiling is 97 % at $\mu = 0.5$ and above 99 % everywhere below $\mu \sim 0.1$, and an ordinary 5σ cut attains it. That is Equation (6) restated: when reads are almost all 0 or 1 electron, all the flux information is in how often a read escaped the read-noise peak, and subdividing what lies above the cut adds nothing. If your science lives entirely below a tenth of an electron per read, eight bins buy you a percent, and you should keep the threshold and the simplicity that comes with it. This is also the regime in which Harpsøe et al. (2012, A&A 537, A50; `arXiv:1111.2066`) found thresholding to recover shot-noise-limited fluxes, "up to fluxes of about one photon per pixel per readout".

Above a few electrons per read, thresholding hits a hard ceiling at $\sqrt{2/\pi}$. The ceiling curve does not decay: it settles onto a plateau near 79 to 80 % and stays there. The number is not a coincidence of this detector. At high flux the pixel value becomes a smooth, effectively location-like family in μ, and reducing any such measurement to a single bit is the classical one-bit quantiser problem, whose Fisher efficiency is maximised by cutting at the median:

$$\eta_{1\,\text{bit}} = \frac{\varphi(0)^2}{\frac{1}{2}\cdot\frac{1}{2}} = \frac{2}{\pi} = 0.6366, \qquad \sqrt{\eta_{1\,\text{bit}}} = 79.8\,\%, \tag{11}$$

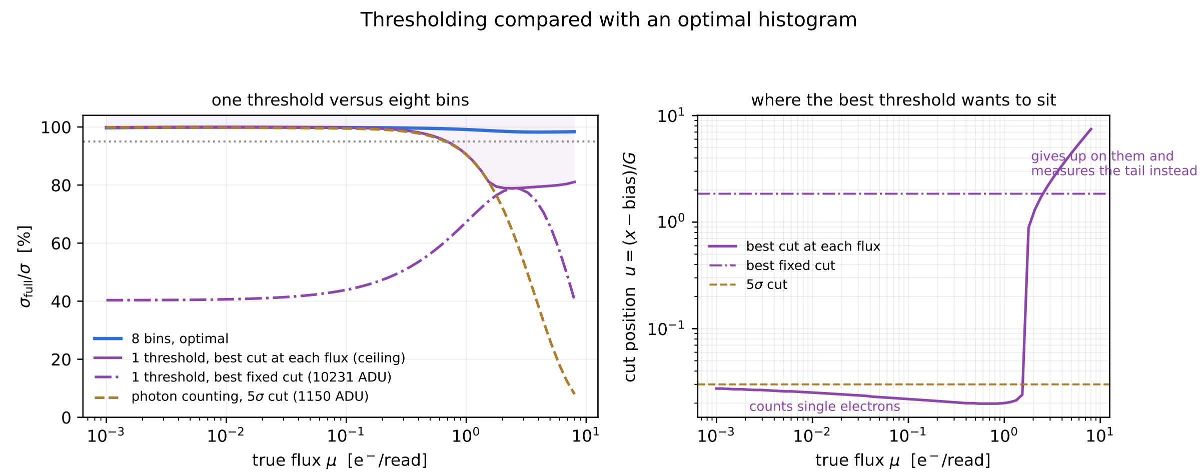

Figure 5: Left: accuracy retained by a single threshold, in its three variants, against the eight-bin design. The shaded band above the ceiling curve is unreachable by any thresholding scheme. Right: where the best threshold wants to sit, in gain units $u = (x - b)/G$. Below $\mu \sim 1$ e$^-$/read it hugs the read-noise peak and counts single electrons; above it, it abandons them and measures the EM tail instead. The two regimes are separated by two decades in u, which is why no one fixed cut serves both.

with φ the standard normal density. Pushing the worked detector's saturation far enough out to reach $\mu = 40$ e$^-$/read gives a numerical best-threshold efficiency of 0.6366, matching (11) to four digits. So a fifth of the precision is not lost to a bad choice of cut; it is the price of the bit itself. One threshold cannot tell one electron from two, and above a few electrons per read that multiplicity *is* the signal.

A threshold asked to serve both regimes serves neither. The best fixed cut over $[10^{-3}, 8]$ e$^-$/read retains 40.4 % in the worst case, which is worse than the ceiling everywhere and far worse than the 5σ cut at low flux. The right-hand panel of Figure 5 shows why: the optimal cut sits at $u \simeq 0.02$, hugging the read-noise peak, while $\mu \lesssim 1$, then jumps by two decades to sit inside the EM tail. There is no compromise position between counting electrons and measuring the tail, so the max-min cut is forced into a gap that is bad on both sides. Eight bins have no such problem: they hold the low cut *and* the ladder at the same time, which is the whole reason the design has more than one number in it.

The practical summary is a boundary rather than a verdict. Thresholding is the right tool below roughly one electron per read and costs almost nothing there; the histogram is the right tool above it, and the crossover is sharp because it is the point at which double electrons become common. Since a real field contains sky at 10^{-2} and star cores at several e$^-$/read in the same frame, and the same reduction must serve both, the histogram is what the instrument needs.

6 Monte-Carlo demonstration

Everything above is a Fisher-information calculation. It predicts the variance of an efficient estimator; it does not by itself prove that a real maximum-likelihood fit to eight bins attains it. Figure 6 and Table 6 close that gap with simulated data.

The procedure, implemented in `demo_optimal_bins.py`, is: draw reads from the exact signal chain (Poisson, then Gamma, then Gaussian, then saturate); bin the *same* simulated reads under each design, so that no Monte-Carlo noise enters the comparison between designs; fit μ by grid maximum likelihood; and measure the scatter over 1200 independent pixels. The number of reads per pixel is set inversely to the flux so that every point rests on a comparable number of detected electrons, roughly 400; with a fixed read count the low-flux estimator would be visibly discrete and its scatter would stop meaning what the Cramér-Rao bound predicts. The reference is a 881-bin histogram whose efficiency against the exact 1 ADU calculation is 0.99999, so it stands in for "keep everything".

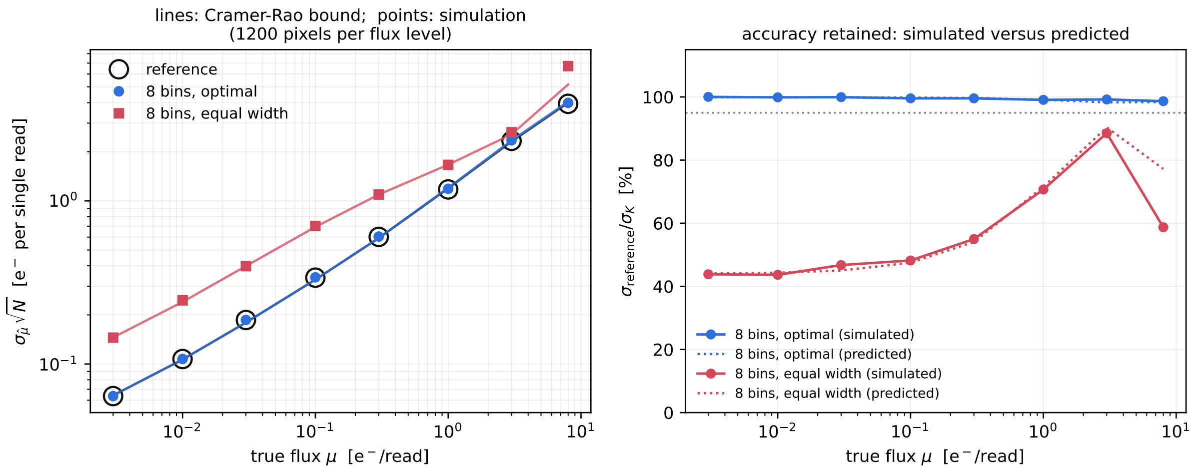

Figure 6: Left: simulated scatter of the flux estimate, normalised to one read, against the Cramér-Rao bound (lines). The eight-bin points sit on the reference curve. Right: accuracy retained, simulated against predicted. The eight-bin design tracks its prediction to better than one percent; equal-width bins track theirs too, at a much worse level.

Table 6: Monte-Carlo results, 1200 independent pixels per flux level. $\sigma_1 = \sigma_{\hat\mu}\sqrt{N}$ is the scatter normalised to a single read, directly comparable to $1/\sqrt{I(\mu)}$.

μ	N reads	σ_1 reference	σ_1 8 bins	accuracy retained	predicted
0.003	133 333	0.0636	0.0635	100.0 %	99.9 %
0.010	40 000	0.1073	0.1074	99.9 %	99.9 %
0.030	13 333	0.1861	0.1862	99.9 %	99.9 %
0.100	4 000	0.3385	0.3400	99.6 %	99.8 %
0.300	2 000	0.6019	0.6045	99.6 %	99.7 %
1.000	2 000	1.1758	1.1866	99.1 %	99.1 %
3.000	2 000	2.3359	2.3542	99.2 %	98.3 %
8.000	2 000	3.9387	3.9916	98.7 %	98.4 %

The agreement is close throughout: the simulated worst case over the eight flux levels is 98.7 % against a predicted 98.3 %, the difference being consistent with the residual Monte-Carlo scatter of the variance estimate (about 2 % for 1200 trials). The claim in the summary box is therefore not a theoretical bound alone; a real fit to eight real histograms delivers it.

Over the same simulations, eight equal-width bins retain only 44 % of the accuracy at low flux, which is the practical cost of not designing the bins.

7 Using the code

`bin_optimizer.py` exposes one general-purpose entry point. Give it the detector and the flux range, and it returns the edges.

```
from bin_optimizer import design_bins

d = design_bins(bias=1000, gain=5000, ron=30, saturation=60000,
                nbins=8, mu_min=1e-3, mu_max=8.0, cic=1e-3)
```

```
print(d.summary()) # every cut, in ADU, in u, and in RON sigmas
d.edge_list() # [0, 1125, 6176, ..., 60000, 60001]
d.worst_accuracy # 0.9827 -> 98.3 % of the full histogram
d.eta # efficiency at each flux in d.mu_grid
```

The same thing from the command line, including a scan over K:

```
python bin_optimizer.py --bias 1000 --gain 5000 --ron 30 --sat 60000 \
                  --nbins 8 --mu-min 1e-3 --mu-max 8 --cic 1e-3 --scan
```

The `method` keyword selects `'dp'` (the max-min dynamic programme), `'closed'` (the interpretable three-zone family, whose fitted parameters are returned in `d.params`), or `'both'`, the default, which computes each and keeps whichever wins. `'closed'` is the one to use when the edges must be explainable in a paper; the difference in worst-case η is under 0.2 % at $K = 8$.

`demo_optimal_bins.py` regenerates every figure and table in this document, including the Monte Carlo, in about a minute.

8 A second detector: large read noise relative to gain

The worked example has $r = \sigma/G = 0.006$: the read-noise peak is a needle compared with the gain, and one cut serves it. It is worth checking a detector where that is not true. The PESTO EMCCD at Mont-Mégantic, whose constants come from the `pesto_stats` MCMC calibration (b = 303.6178 ADU, G = 71.060 ADU/e$^-$, σ = 6.8164 ADU, saturation at 5000 ADU), has $r = 0.0959$ and $S = 66.1$, and is used here over $\mu \in [10^{-3}, 4]$ e$^-$/frame. At $K = 8$:

bin	lower [ADU]	upper [ADU]	u_{lo}	u_{hi}
0	0	320	−4.273	0.231
1	320	331	0.231	0.385
2	331	404	0.385	1.412
3	404	492	1.412	2.651
4	492	601	2.651	4.184
5	601	740	4.184	6.140
6	740	949	6.140	9.081
7	949	∞	9.081	∞

Worst-case $\eta = 0.9618$, accuracy 98.1 %: eight bins again suffice. But the design has adapted in two ways that the physics of Section 3 predicts.

First, there are now *two* cuts through the peak, at $+2.40\sigma$ and $+4.02\sigma$, rather than one. Table 2 shows why: at $r = 0.0959$ a single cut at 4σ would dump 32 % of all single electrons into bin 0, and a cut low enough to avoid that would leak read noise. When neither leak can be made small, the only remedy is to spend a second bin resolving the overlap region instead of discarding it. The rule of thumb that emerges is that one cut suffices while $c_1 r \ll 1$, and a second is worth its bin once $c_1 r \gtrsim 0.2$.

Second, the top cut sits at $u = 9.08$, far below the saturation head-room $S = 66.1$, and close to $\mu_{\max} + 2.5\sqrt{\mu_{\max}} = 9.0$ for $\mu_{\max} = 4$. With generous head-room, placing the last edge at the physical saturation level would waste it on a region no read ever reaches.

8.1 The design this repository ships

Memory is not the binding constraint on this instrument – a $16 \times 426 \times 1024$ cube of counts is 28 MB per chunk of frames – so `embin_config.yaml` ships the $K = 16$ design rather than the $K = 8$ one, and buys back most of the remaining loss:

K	4	8	12	16	24	32
η	0.8683	0.9618	0.9815	**0.9887**	0.9940	0.9959
accuracy	93.2 %	98.1 %	99.1 %	**99.4 %**	99.7 %	99.8 %

The edges, in the convention of `embin.py` where the final entry is a formal sentinel for the open top bin:

```
edges: [0, 314, 321, 328, 332, 360, 397, 444, 500, 565,
        640, 725, 819, 922, 1034, 1156, 1157]
```

The closed-form family of Section 4 wins here, with $m = 3$ peak cuts, $c_{\mathrm{lo}} = 1.5$, $c_{\mathrm{hi}} = 3.5$, $u_1 = 0.4$ and $u_{\mathrm{top}} = 12.0$: four edges fall inside the read-noise peak, at $+1.52$, $+2.55$, $+3.58$ and $+4.16\sigma$, and the eleven rungs above it are spaced uniformly in $\sqrt{u}$ to within 1.6 % (increments 0.255 to 0.259). Efficiency is flat at $\eta = 0.989$ across the whole flux range, against a 16-bin equal-width design which retains only 60 % of the accuracy at the sky level of this instrument.

8.2 On real frames

Applied to a 1001-frame PESTO sequence (48.97 ms per frame), cut into fifteen chunks of 64 frames so that the field's 0.16 pix s^{-1} drift stays below half a pixel within a chunk, the fitted sky flux is 0.0159 to 0.0170 e^-/frame chunk to chunk, against 0.0162 ± 0.0001 from the independent MCMC calibration of the same data: the binning introduces no detectable bias. Of the 2.79×10^7 pixel values in a chunk, 39 land in the open top bin, which is the top cut doing exactly what Section 3 asks of it – sitting where the counts stop, and not one bin higher.

9 Scope and caveats

- The design is optimised for the *worst case* over a stated flux range. Widening that range costs accuracy at every flux, so it should be set to what the science needs and no wider. Optimising to $\mu_{\max} = 1$ instead of 8 raises the worst-case η at $K = 8$ appreciably.
- Efficiency is measured against the full 1 ADU histogram, which is the correct ceiling for a histogram-based method. It is not a comparison against an ideal photon counter: the EM register's own excess noise costs information that no binning can recover, and is present in both terms of the ratio.
- The bias, gain and read noise are treated as known. They come from a separate calibration; if they are fitted jointly with the flux, the relevant object is a Fisher *matrix* and the optimal bins shift, generally towards resolving the peak more finely, since that is where the bias and read noise are constrained.
- Bin edges are whole ADU, because the detector is digitised. The cuts through the read-noise peak are rounded *up*, never down: rounding a cut down towards the bias admits read-noise leakage, and Section 3.2 shows how expensive that is. For a detector whose read noise is smaller than one ADU the rounding direction alone can change η from 0.99 to 0.51.
- Clock-induced charge is modelled as an additive Poisson term and is not separable from the flux by a single-pixel histogram. The estimator returns μ with μ_{CIC} already subtracted, on the assumption that μ_{CIC} is known.

References: Basden, A. G., Haniff, C. A., & Mackay, C. D. 2003, MNRAS, 345, 985. Harpsøe, K. B. W., Andersen, M. I., & Kjægaard, P. 2012, A&A, 537, A50. Daigle, O., Carignan, C., Gach, J.-L., et al. 2009, PASP, 121, 866.